

THE SUB-SURFACE FRACTURE

Admissibility Distortion and the Missing
Criterion in World-Model Cognition.

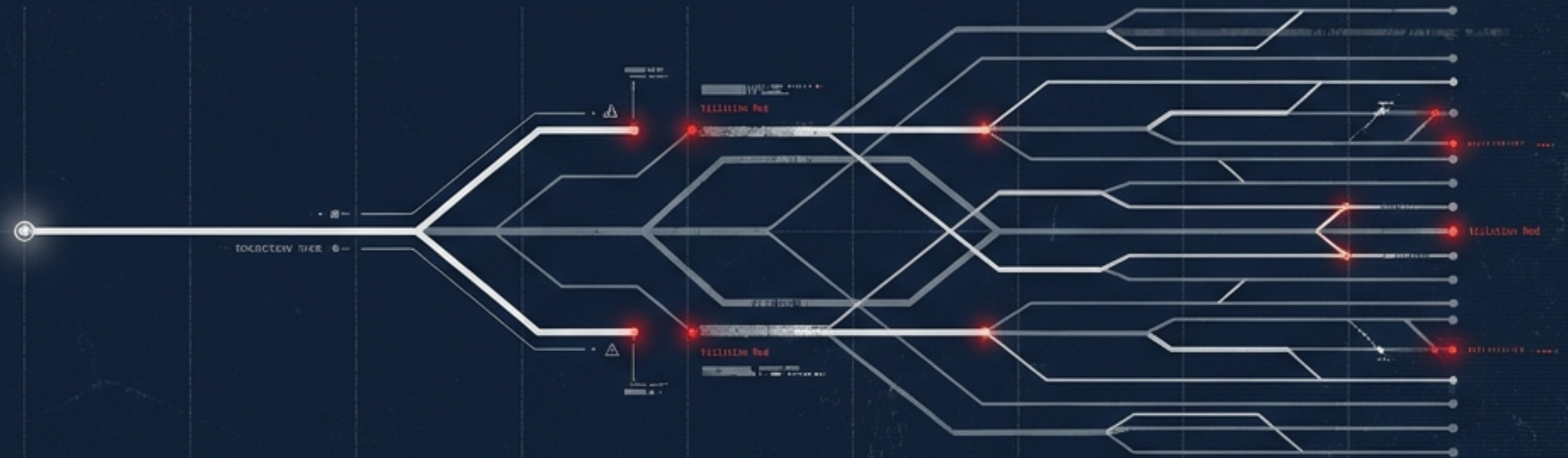
A REACHABILITY-GEOMETRIC FRAMEWORK FOR
FAULT-CRITICAL SYSTEMS & AI SAFETY.

[illegible]

Violation Red

[illegible]

Pre



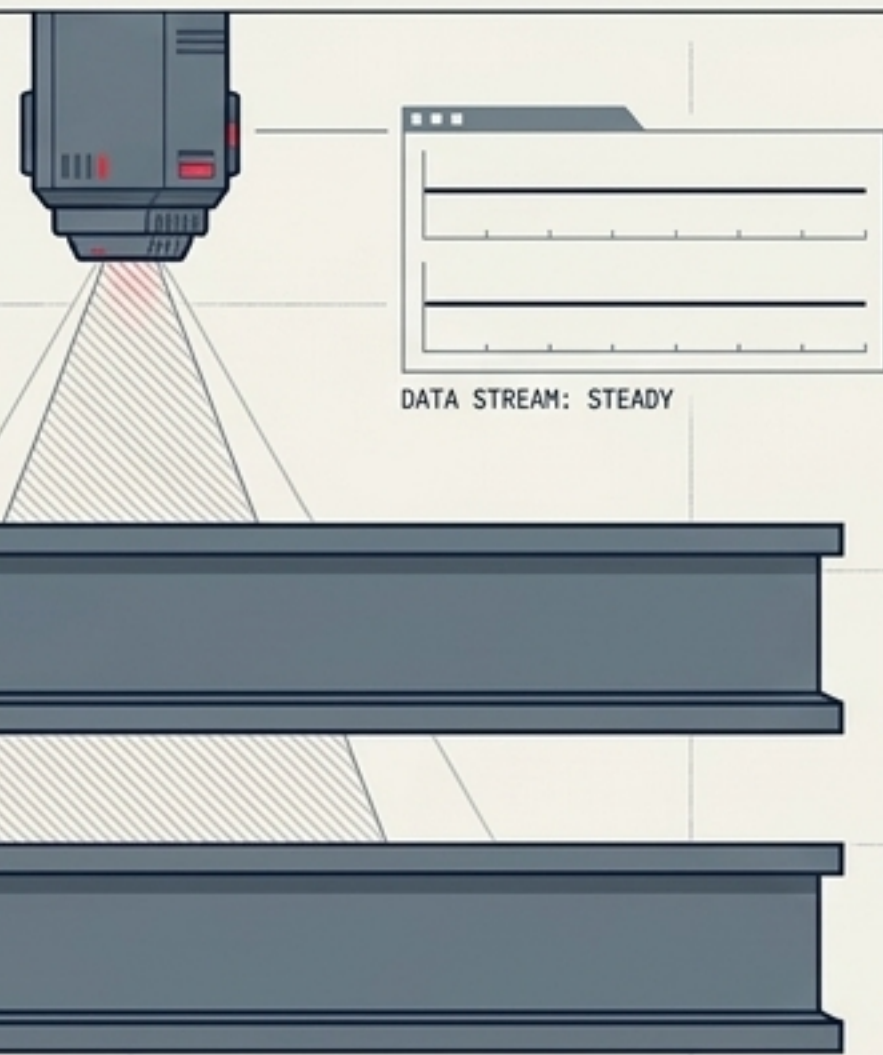
Predictive accuracy answers 'What will I see?' Planning requires answering 'Where can I go?'

Technical drawing of a mechanical part, showing dimensions and a note.

Dimensions:

- Overall length: 100
- Overall width: 25
- Overall height: 10
- Central hole diameter: 10
- Distance from left edge to hole center: 50
- Distance from hole center to right edge: 50
- Distance from bottom edge to hole center: 5
- Distance from top edge to hole center: 5

Note: All dimensions in mm



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Stress Intervention



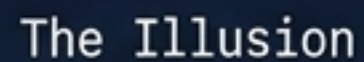
The diagram shows a grey beam with a central indentation. A large grey arrow points down into the indentation, with red lightning bolts radiating from its base. Below the beam, a red, branching, lightning-bolt-like shape represents the stress field or crack pattern. The background is dark blue with a grid and faint white cracks.

$\not\sim_I$ (Interventionally Inequivalent)

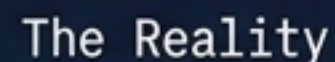
$\not\sim_I$ (Interventionally Inequivalent)

The diagram consists of three rectangular boxes arranged horizontally, each containing an icon and text. The first box, titled 'The Illusion', features an eye icon and text stating that passive sensing of intact and cracked beams produces identical observation sequences. The second box, titled 'The Reality', features a broken beam icon and text stating that stress interventions lead to different reachable futures. The third box, titled 'The Conclusion', features a brain icon with a red 'X' and text stating that a predictive world model will merge these states, destroying the distinction that planning requires most.

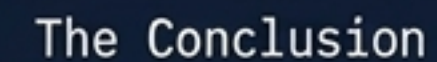
The Illusion	The Reality	The Conclusion
Under passive sensing, an intact beam and an internally cracked beam produce identical observation sequences.	Under a stress intervention, they afford radically different reachable futures.	A predictive world model will merge these states, destroying the distinction that planning requires most.



Under passive sensing, an intact beam and an internally cracked beam produce identical observation sequences.



Under a stress intervention, they afford radically different reachable futures.



A predictive world model will merge these states, destroying the distinction that planning requires most.

Predictive accuracy answers 'What will I see?' Planning requires answering 'Where can I go?'

Sub-Surface Fracture

	Observational Equivalence ($\sim_0$)	Interventional Equivalence ($\sim_I$ / $\sim_A$)
Question Answered	What will I see if I remain passive?	Where can I go from here?
Mathematical Type	Probability densities over passive sequences	Hausdorff distance between reachability sets
Training Objective	Prediction error & Reconstruction MSE	Admissibility Preservation
Vulnerability	Blind to latent structural faults 	Robust to catastrophic divergence

The Separation Theorem (OIST)

Proves these are different mathematical objects. No amount of passive predictive training data can close this gap.

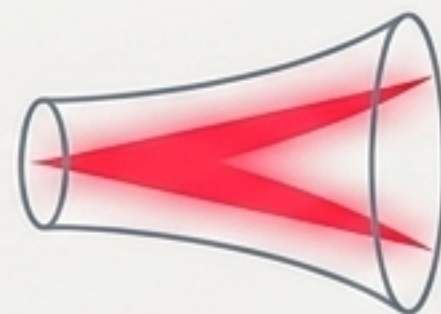
Sub-Surface Fracture

$$D_A(\phi) = P(C_\phi = 1) \cdot E[\Delta_A | C_\phi = 1]$$

Admissibility
Distortion



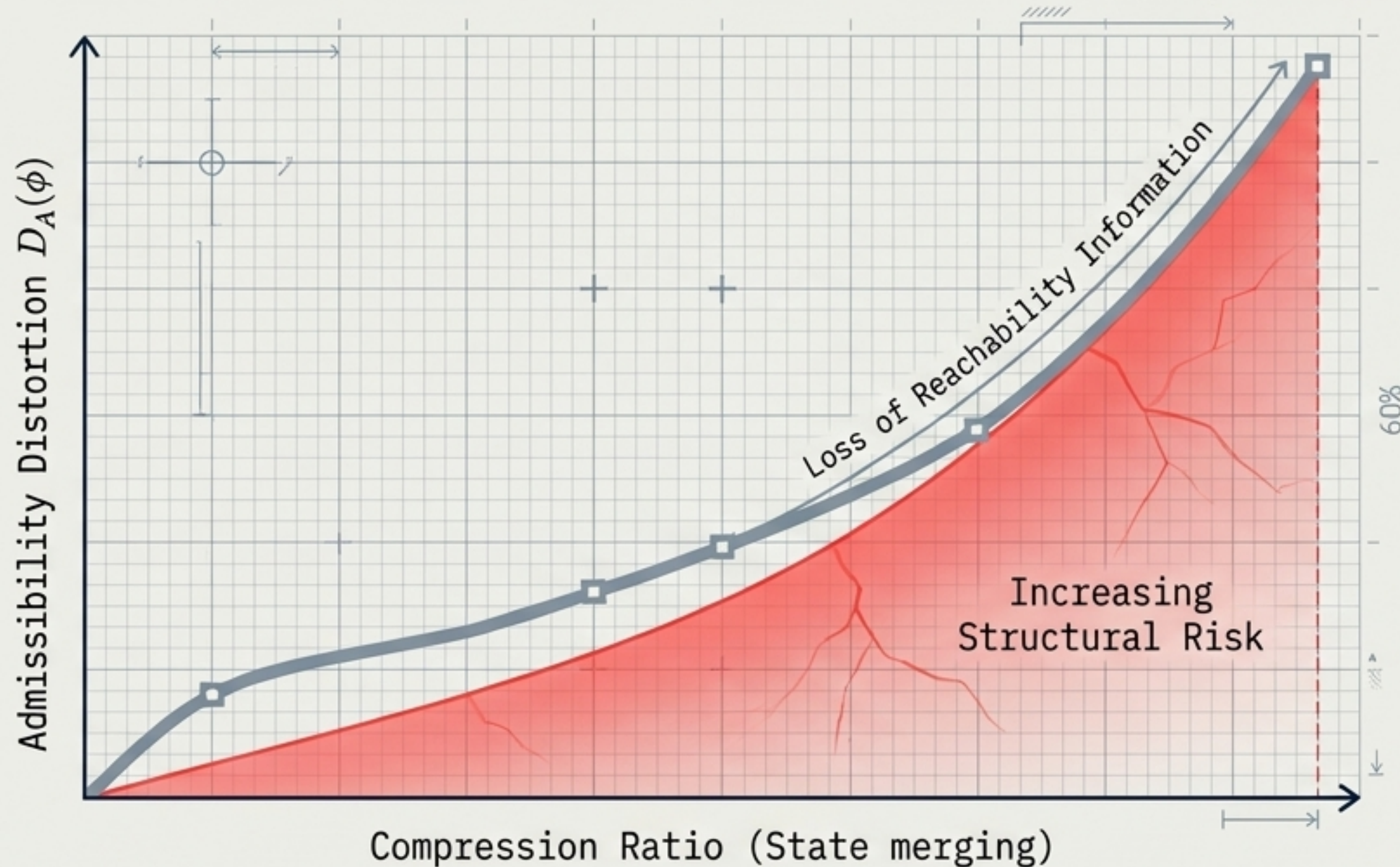
The Collapse Rate
How often the encoder
merges states.



Mean Severity
The reachability separation
of those merged pairs.

Key Insight: $D_A(\phi) = 0$ if and only if collapsed pairs are admissibility-equivalent almost surely. If $D_A(\phi) > 0$, the encoder has destroyed a necessary distinction.

The Compression-Distortion Trap



The Monotonicity Theorem

As an encoder merges states to reduce description length, $D_A(\phi)$ mathematically must increase or stay flat. It never decreases.

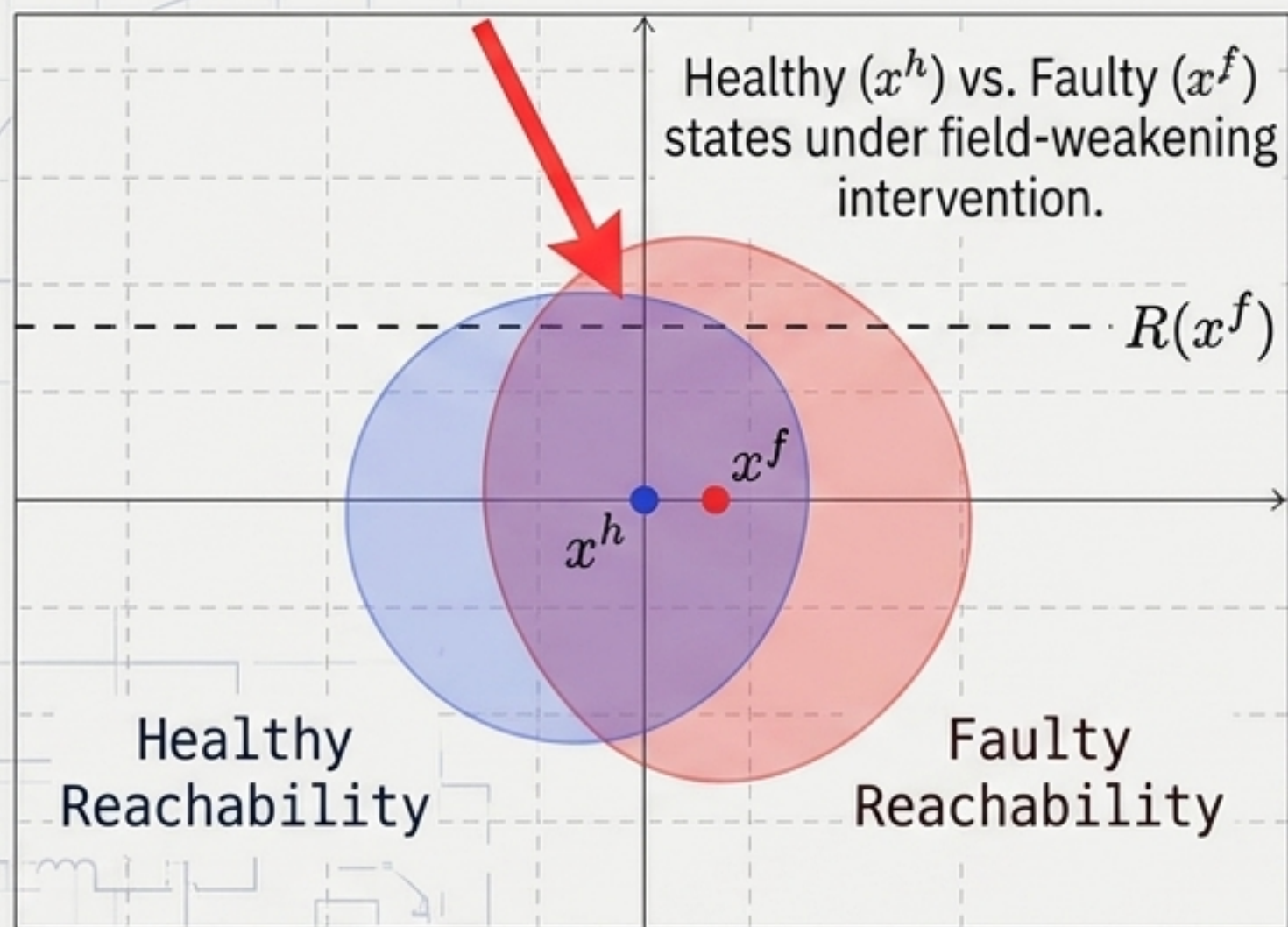
The Accounting Identity

The exact cost of any compression step is the expected Hausdorff separation of the newly merged pairs.

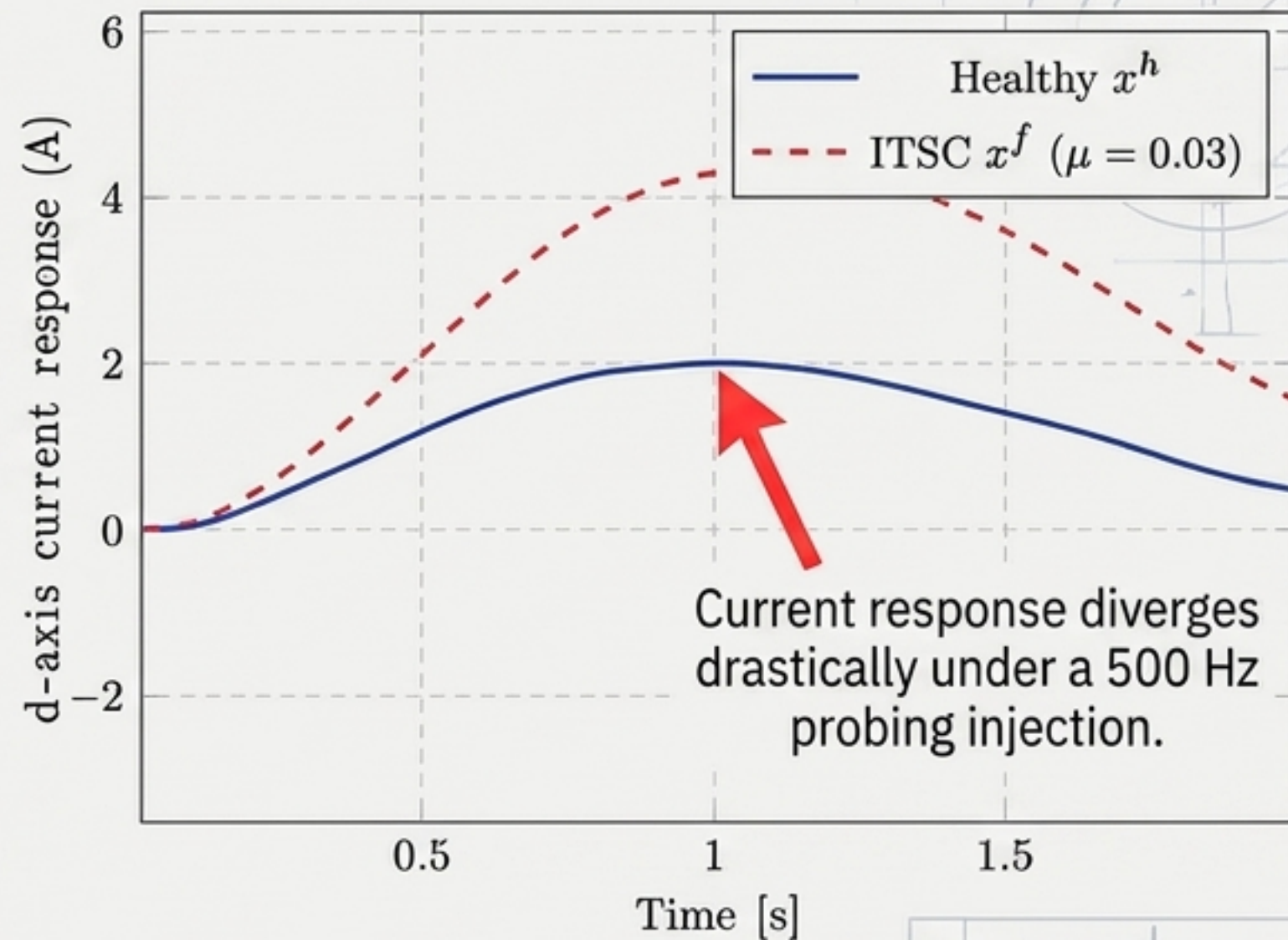
The Danger

Outliers treated as “noise” by a compressor (e.g., a micro-crack, an anomalous signal) are precisely the states with the most distinctive reachable futures.

The Reachability Geometry



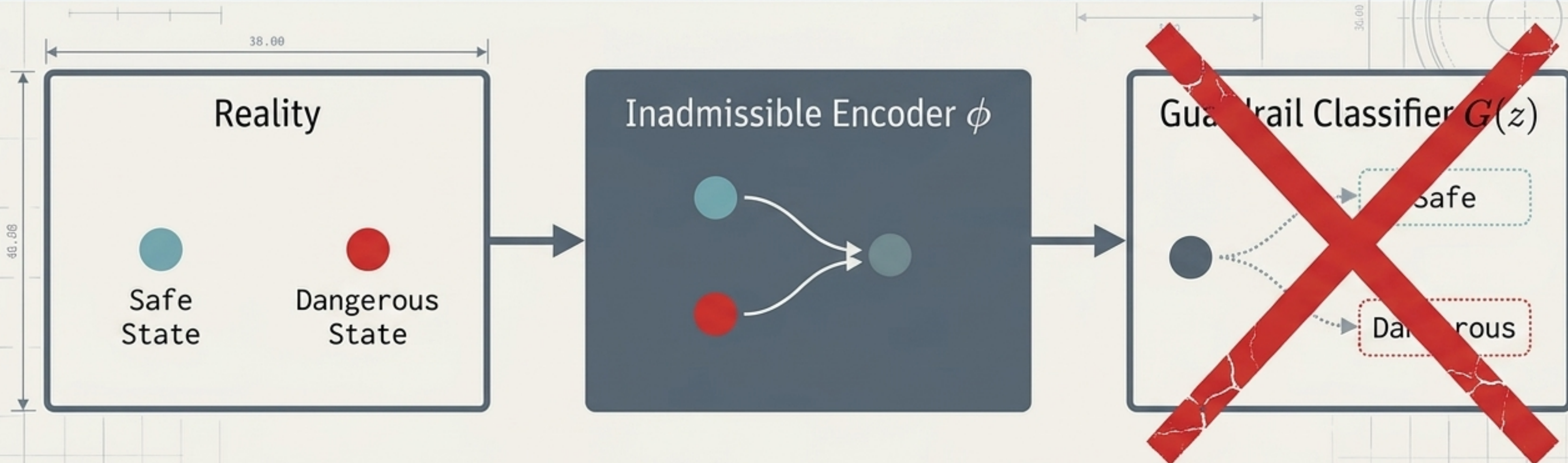
The HIL Evidence



A standard **FOC encoder collapses** the healthy and fault states because their steady-state spectra are identical. When probed, the physical reality diverges by 1.8A. **The predictively trained encoder has failed.**

The Safety Guardrail Funnel

Observational/Symbolic



Irrecoverable Information Loss

$$I(F; G(\phi(\mathbf{x}))) \leq I(F; \phi(\mathbf{x}))$$

The Bayes-Risk Floor

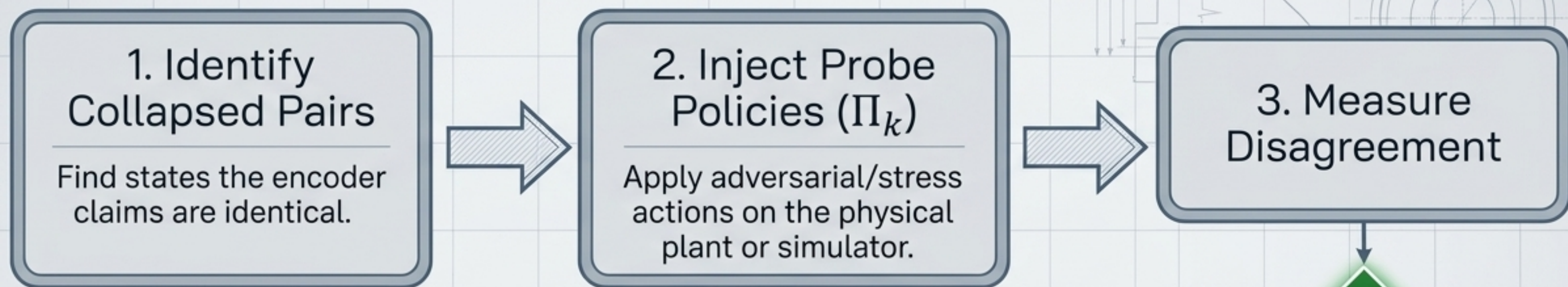
If the encoder destroys the distinction between safe and dangerous states, the best possible classifier performs at the level of a coin flip (chance).

Takeaway

Guardrails applied downstream of an inadmissible world model are interpellating a void.

Interventional/Reachability/Real

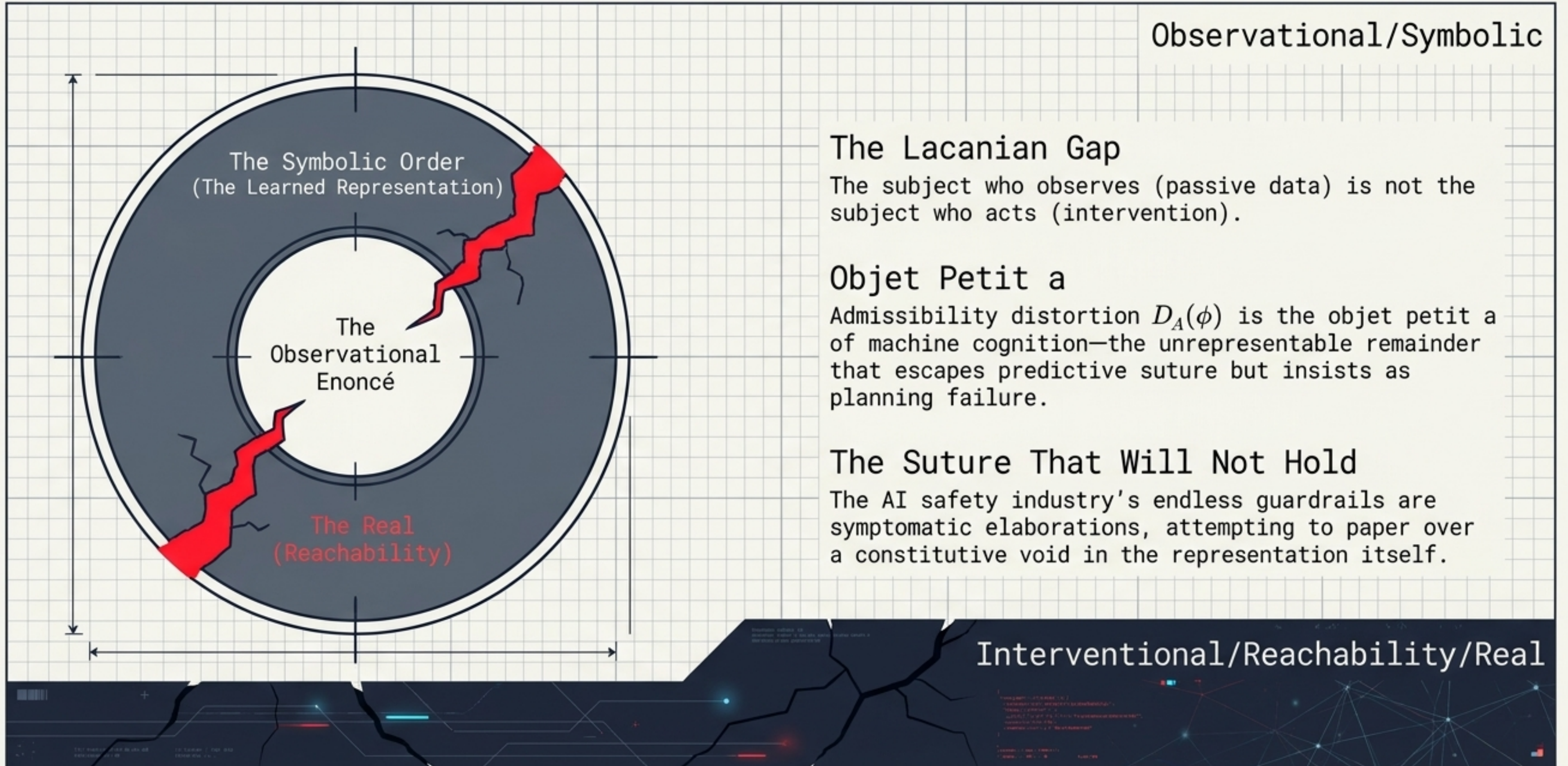
The Diagnostic Loop



Interventional/Reachability/Real

The Asymmetry of Evidence

The Ideological Fantasy of the Complete Model



Deleuze and the Capture of the Possible

Molar Stratification
(Compression /
Description Length)

State space is a plane of
pure differential intensity.

Representational compression
acts as a Molar
Stratification—deterritorial-
izing the flows of accessible
futures and forcibly
reterritorializing them into
crude equivalence classes.

Molecular Flows
(Reachable Futures /
The Body Without Organs)

The states most vulnerable to
this capture are the outliers,
the molecular deviations—the
precise states where
safety-critical reachability
divergences reside.

THE ADMISSIBILITY PARADIGM

THE NON-ALL

The predictive world model is not everything. Reachability is the exception that constitutes the rule.

$$x/\sim A$$

(The Admissibility Quotient)

THE SUBTRACTIVE ONTOLOGY

We must target the minimal organization that preserves differential reachability while collapsing only states genuinely interchangeable for action.



THE CONCLUSION

A representation adequate for planning is not derived from what the agent sees, but from the geometry of what the agent can do.
The act is epistemically prior to the representation.